

Bitmap Indices

Review: Bitmap / Bitset

Set of Booleans represented as sequence of bits

- Often encoded in a byte or word

{0: T, 1: T, 2: F, 3: T, 4: F, 5: T}

= 110101 (least significant bit first) = 53

= 101011 (most significant bit first) = 43

Basic idea of bitmap indices

Assume tuples ordered 0, 1, …

Record #	id	gender	income_level
0	51351	M	1
1	73864	F	2
2	13428	F	1
3	53718	M	4
4	83923	F	3

Bitmaps

M	F	IL1	IL2	IL3	IL4	IL5
1	0	1	0	0	0	0
0	1	0	1	0	0	0
0	1	1	0	0	0	0
1	0	0	0	0	1	0
0	1	0	0	1	0	0

Stored as columns

Queries use standard bitmap operations

```
SELECT … FROM … WHERE gender = 'F' AND income_level = 1;
```

F's bitmap 01101 & 1's bitmap 10100 = 00100

```
SELECT … FROM … WHERE gender = 'F' OR  income_level = 1;
```

F's bitmap 01101 | 1's bitmap 10100 = 11101

```
SELECT … FROM … WHERE income_level <> 1;
```

~ 1's bitmap 10100 = 01011

Space overhead

Normal: #tuples × (#values + 1) bits, if nullable

- Typically used when # of values for attribute is low.

Encoded: #tuples × log(#values)

- But need to use more bitmaps

Compressed: ~50% of normal

- Can do bitmap operations on compressed form.

A couple details

When deleting tuples, either …

- Delete bit from every bitmap for that table, or
- Have existence bitmap indicate whether that row exists.

Can combine bitmaps with B+-trees.

- Normal: B+-tree leaf holds list of pointers to tuples with a particular attribute value.
- Combo: Can instead hold bitmap for that value.